METAMATCH: RECONCILING METADATA FOR AGGREGATING MUSIC DIFFICULTY ANNOTATIONS

Megan E. Finch[1] **Chris G. Willcocks**[1] **Mark R. H. Gotham**[2]
[1] Durham University, UK [2] King's College London, UK
{megan.e.finch,christopher.g.willcocks}@durham.ac.uk Mark.Gotham@kcl.ac.uk

ABSTRACT

The inconsistency of metadata across sources is a persistent problem for large-scale music research. The variety of approaches to documenting even basic elements such as artists' names and work titles makes it difficult to aggregate data across different sources. This paper presents a comparative analysis of approaches to this problem, combining the OpenRefine Reconciliation pipeline with LLM-based strategies for entity matching. We also explore one potential application of this pipeline in sheet music difficulty classification. By gathering difficulty labels from established music exam board syllabi and matching them to corresponding IMSLP entries, we link expert difficulty annotations to sheet music available through IMSLP for automatic sheet music difficulty classification. The resulting piano dataset consists of 7,737 repertoire-list entries, of which 2,389 are automatically matched to 1,210 distinct IMSLP work identifiers. Our method produces data that combine high-quality expert annotations with the scale more typically associated with crowdsourced corpora, and can be extended to additional instruments. We release the matching pipeline, dataset, and code at `https://meganelisabethfinch.github.io/metamatch` to support further work on cross-corpus music analysis, both for this use case and for the many others which depend on robust metadata alignment at scale.

1. INTRODUCTION

Music information retrieval research increasingly draws on diverse, independently-curated, and specialised music datasets, ranging from multimodal music question-answering datasets [1] to large-scale music encyclopedias [2]. However, the heterogeneity of data schemas used and the lack of standardised metadata mean that cross-collection search and analysis remain challenging.

As a motivating example, recent progress in the automatic estimation of music performance difficulty from sheet music images has predominantly focused on piano music [3–6]. Research on other instruments is limited by a scarcity of difficulty-labelled sheet music data. That is not for want of rich sources of expert-quality difficulty labels across multiple instruments: music exam board repertoire lists collectively grade tens of thousands of works across many instruments. Moreover, many of these works are also in the public domain and, as such, have sheet music available in free online repositories such as IMSLP [1] and PDMX [7]. The limitation is instead at the metadata level: to make use of this data, researchers must first link musical works across multiple sources.

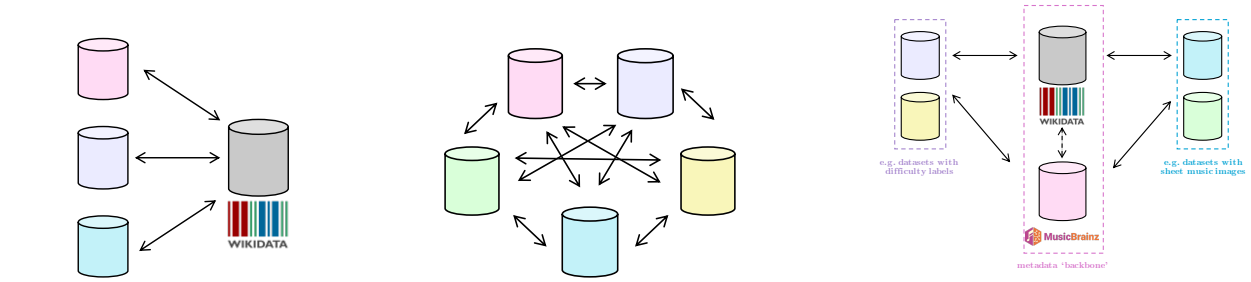
Existing approaches to cross-corpus linking, such as the LinkedMusic project [8, 9], make use of tools such as OpenRefine's Reconciliation API [2] to semi-automatically match and link values to their corresponding Wikidata Uniform Resource Identifiers (URIs). Cross-collection queries can then be formed by joining datasets on these reconciled URIs. In practice, however, this approach often fails to find many of the matches needed for a full join. For example, many musical works (and even many artists) have no corresponding Wikidata entry, and, while there exist more music-specific knowledge bases [2], no single database covers all the musical entities one might wish to link. The alternative approach of direct, 'peer-to-peer' reconciliation of music datasets with each other is also limited, in part because it scales poorly with the number and size of datasets, and more importantly because effective reconciliation relies on the alias lists provided by large-scale data 'backbones', such as Wikidata. Figure 1 sets out some of these approaches schematically.

Moreover, most automatic reconciliation APIs treat entity matching as a binary classification problem: their aim is to determine whether two records describe the same entity or not. However, under different schemas, a 'musical work' entity might describe an entire album, its constituent tracks, or even multi-author compilations. In classical music, further variation arises from the splitting (or not) of 'movements', such as separating a recitative from an aria or the constituent elements in a set of 'variations'. Figure 2 provides an example of divergence in the handling of a single-author collection of pedagogical pieces relevant to the present use case.

Overall, effective linking across music datasets at scale requires a reconciliation stage that is 1) not beholden to the reliability of any single 'backbone', and 2) also capable of

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[1] `https://imslp.org/`
[2] `https://openrefine.org/`



(a) LinkedMusic (single 'backbone') (b) Match between pairs of datasets (c) Ours (multiple 'backbones')

Figure 1: LinkedMusic (1a) reconciles entities in each dataset against Wikidata, then uses the shared Wikidata identifiers to bridge between datasets; this means that entities not matched to Wikidata cannot be connected by queries. Directly matching between music datasets (1b) is challenging due to inconsistent metadata, and does not scale well to many datasets. Our approach (1c) attempts to find a more complete matching by routing through one of several 'backbone' datasets (e.g., Wikidata or MusicBrainz), and by leveraging the existing links between them as a weak supervision signal.

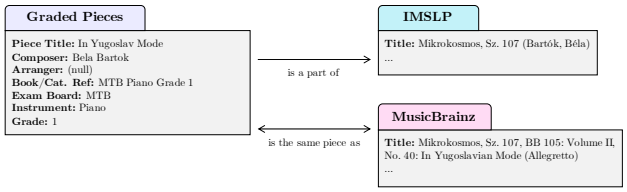


Figure 2: An example of the possible relationships between representations of the same 'musical work' across different datasets.

expressing more nuance in the relationships between entities. In this paper, we propose a solution on that basis.

2. METHODOLOGY

Our approach to entity matching involved matching entities across datasets by routing through multiple data 'backbones'. Section 2.1 introduces the formal framing; Section 2.2 describes our use of LLMs to match pieces; and Section 2.3 integrates the LLM-based matching into a complete pipeline to create a new difficulty-labelled sheet music dataset for piano.

2.1 Problem Formulation

We frame cross-corpus music linking as a matching problem. Given a set of music datasets $\mathcal{D} = \{D_1, D_2, \ldots, D_n\}$, each dataset $D \in \mathcal{D}$ contributes a set of entities, written $x_{D,i}$ for the ith entity in D.

Our task, given a query dataset D_q and a target dataset D_t, is to classify candidate pairs with a relation $r \in \{\texttt{exact}, \texttt{part of}, \texttt{contains}, \texttt{no match}\}$. Several entries may link to the same work, so this need not be one-to-one. Compatible relations may also be composed through intermediate datasets along a path $D_q \to \ldots \to D_i \to \ldots \to D_t$.

In this work, we use large metadata backbones as intermediate datasets to connect query and target. Here we use Wikidata and MusicBrainz (hereafter WD and MB, respectively), while noting that any valid path suffices for this task: different matched pairs may be routed through different backbones, or matched directly without an intermediate. The remainder of this section describes how we use LLM-based classifiers to infer the edges of this matching.

2.2 LLM-based Entity Matching

The crux of the reconciliation process is to identify separate instances of the same entity (e.g., a musical work) across databases. Simple string matching or regular expression-based baselines are generally insufficient here, due to the extensive variations routinely encountered across spelling (e.g., *The Duke of Glocester's March* vs the alternative spelling of *Gloucester*), translation (e.g., *Pictures at an Exhibition* vs *Tableaux d'une Exposition*), and variable detail in reporting on constituent parts of wider compositions or collections (e.g., *Bydlo* vs *Pictures at an Exhibition*). Figure 4 (below) sets out some of these and other variations as relevant to the present use case.

These challenges clearly motivate the use of LLMs for the task. We leveraged LLMs to identify relations between pieces, with the goal of determining, for two pieces x and y, whether these are distinct compositions, the same composition, one is a 'part' of the other, or one 'contains' the other.

Prior work has shown that LLMs with little to no training perform comparably to pre-trained language models which have been fine-tuned for the entity matching task [10]. Therefore, we performed few-shot prompting and asked the model to classify the pair, given a pair of piece descriptors formed from the metadata of each entry (i.e., the piece title, composer, and book).

2.3 Graded Pieces Dataset

We introduce a new dataset of music piece metadata with expert difficulty labels, compiled from the performance grade repertoire lists of five music exam boards; in total, this comprises **58,104** difficulty-labelled entries for over 20 instruments. The grade assignments were transcribed from published syllabi of multiple exam boards.

To illustrate a concrete use case of our reconciliation strategy, we present an automatic matching to IMSLP

works for a subset of this data, as well as a manually curated and labelled subset of 'challenging' cases for evaluation purposes.

2.3.1 Collection

The first task in creating the dataset was collecting graded piece entries, by parsing semi-structured repertoire list PDFs into a structured CSV format, for five exam boards: ABRSM, Trinity College London, Music Teachers' Board (MTB), AMEB, and RCM. For each entry, we collected the piece title, composer and/or arranger, the book or publication in which that piece can be found, the instrument, and the difficulty level, preserving the original grade schemes used by each exam board (e.g., Grades 1–8 for ABRSM, or Levels 1–10 for RCM). [3] This process resulted in **58,104** pieces for over 20 instruments from **13,583** unique composer values. [4]

Having amassed this data, we turn to the subset of works for **piano** (#entries: **7,737**). To this we applied our reconciliation process (as described in Section 2.2) to produce an automated matching between the graded pieces and IMSLP, to yield a dataset consisting of expert difficulty labels paired to the sheet music images in IMSLP's collection.

2.3.2 Composer Reconciliation

Due to the size of the datasets involved, performing pairwise LLM-based classification between all graded pieces and all possible candidates would be prohibitively expensive (e.g., $7,737$ piano entries $\times$ $257,350$ IMSLP works $\approx$ 2 billion combinations). To make this tractable, it was necessary to first filter the set of candidates to search for each graded piece; we did this by first reconciling *other* entities for which the reconciliation task is simpler and better supported.

Therefore, we first matched the *composers* (of both graded pieces and IMSLP) independently to both WD entities and to MB Artist entities. This was performed by OpenRefine's existing Reconciliation API along with some basic filters, such as checking whether the candidate has a 'music-related occupation', to flag potential false positives. Though some MB Artists are already annotated (by human contributors) with their corresponding WD and IMSLP entries, we apply the matching pipeline to each target dataset independently, and leverage these existing links as a weak supervision signal.

Previous work has shown that mutual agreement among independent models or annotators can function as an indicator of whether a prediction is confident (if agreement is high) or not (if low) [11, 12]. Following this principle, we treat composers that can be reconciled to *consistent* WD and MB entities as 'probably correct' matches to compute a formative measure of matching accuracy during development. As shown in Table 1, 48.8% of GP composer values automatically match to a WD composer and 82.2% to an MB Artist. Of the 6,116 GP composer values that match to *both*, 4,179 possess an existing MB→WD link; among these, 3,615 (86.5%) agree and 564 (13.5%) disagree. A high agreement rate here generally indicates a reliable composer reconciliation process. However, we note that this can only be assessed among those composers who both match in MB and WD and have an existing, manually-labelled link from MB to WD. In practical terms, this tends to correspond to better-known composers. Thus, this agreement rate does not necessarily indicate that the composer reconciliation is universally reliable.

	GP.Composers		**IMSLP.Composers**	
Size	13,583		53,564	
Matched to WD	6,624	(48.8%)	31,335	(58.5%)
Matched to MB	11,169	(82.2%)	37,042	(69.2%)
Matched to both	6,116		24,479	
No MB→WD link	1,937		9,300	
Link agrees	3,615	(86.5%)	11,505	(75.8%)
Link disagrees	564	(13.5%)	3,674	(24.2%)

Table 1: Composer reconciliation coverage and MB→WD link consistency, for graded pieces (GP) and IMSLP composer values.

2.3.3 LLM-based Work Reconciliation

After matching composers, we then matched pieces (or 'works') across datasets using LLM-based entity matching, connecting each graded piece to the corresponding IMSLP and MB work(s). For each graded piece, we filtered the IMSLP and MB datasets for candidate works with a matching composer. This generally reduces the candidate pool to a tractable size; most composers listed on IMSLP and MB are associated with $\leq$ 30 works (94.6% and 92.5%, respectively). However, there are nonetheless many prolific composers (e.g., #IMSLP works by Graupner: 1769; by Bach: 1432) for whom LLM classification is still prohibitively slow.

As an additional tractability measure, the remaining candidates are ranked according to the cosine similarity between embeddings of the query entry and each candidate work. The top k candidates per query are then forwarded on to LLM-based matching. This embedding model of all-MiniLM-L6-v2 [13] and candidate limit of $k = 30$ were selected based on performance (i.e., recall@k) on a formative version of the evaluation set. A fast response time was also a priority in selecting this model.

Of the 7,737 piano graded pieces, composer reconciliation succeeded for **5,057** entries. These were passed to the LLM-based matching pipeline to produce a matching against both MB and IMSLP. Table 2 shows the number of matches found by the LLM, disaggregated by type.

As shown in Table 2, many of the 5,057 graded pieces have a predicted match in MB (65.3%) and IMSLP (47.2%). The match rate to MB is notably higher, which is as expected: MB, as a general-purpose open music encyclopaedia, has a larger catalogue of music than IMSLP, which lists only the sheet music it has in its holdings. Similarly, there are many more 'exact' matches against MB (46.1%) than against IMSLP (18.9% exact, 28.2% part of).

[3] Full URLs for sources are provided on the repository at `https://meganelisabethfinch.github.io/metamatch`

[4] Note this reports unique composer *values*, and not unique composers.

	MB.Works		IMSLP.Works	
No match	1,753	(34.7%)	2,668	(52.8%)
Any match	3,304	(65.3%)	2,389	(47.2%)
Exact	2,331	(46.1%)	954	(18.9%)
Part of	782	(15.5%)	1,427	(28.2%)
Contains	191	(3.8%)	8	(0.2%)

Table 2: Match outcomes for the 5,057 composer-reconciled piano graded pieces, against MB and IMSLP.

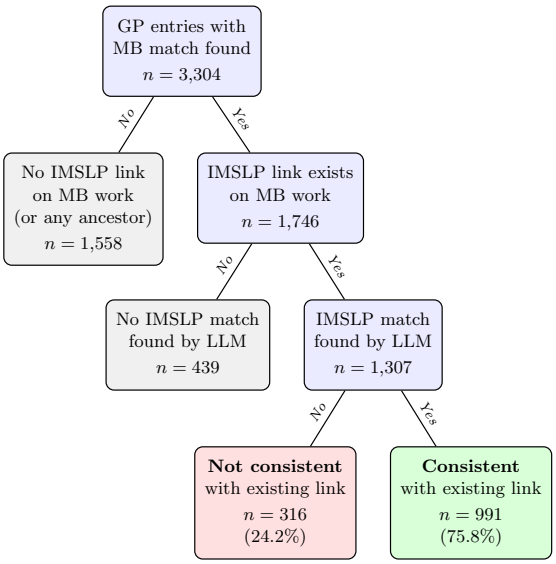


Figure 3: Consistency of LLM-based IMSLP matches against existing MB→IMSLP links, for the 3,304 graded piece entries with a MB match. Of the 1,307 entries with the information needed to check consistency, 991 (75.8%) agree with the existing link.

This is, again, as expected, and emerges as an artefact of how IMSLP is structured: while MB catalogues each individual volume, piece, and movement (e.g., *Mikrokosmos, Sz. 107, BB 105: Volume I, No. 15: Village Song*), IMSLP typically possesses only a single 'work' entry for the entire collection (e.g., *Mikrokosmos, Sz. 107*). Finally, we note that multiple graded piece entries may match to a single MB work—when a piece is listed by more than one exam board, for instance. Therefore, 2,579 distinct MB and 1,210 distinct IMSLP work identifiers are paired with graded piece entries as a result of this process.

Some MB works include existing links to corresponding IMSLP works. As in Section 2.3.2, running the matching pipeline independently against both MB and IMSLP allows us to leverage consistency with existing links as a weak supervision signal of the reliability of the LLM-based matching (see the flowchart in Figure 3). Again, we find that the LLM-derived IMSLP match tends to be consistent with the existing MB→IMSLP link (75.8%).

As a result, we obtain 2,389 predicted (IMSLP URL, difficulty-label) pairs for piano, covering 1,210 distinct IMSLP work identifiers. This is comparable in number of labelled entries to *PianoStreet-Difficulty* (2,816) and *FreeScores-Difficulty* (4,193) [4], while using expert (rather than crowd-sourced) annotations.

All full-dataset work matches reported in this section were produced using Qwen3.5-9B [14]; future work would ideally re-run this matching with a stronger model to yield more reliable matches. Nonetheless, this illustrates how LLMs can be used to construct large-scale datasets linking difficulty annotations to sheet music available through IMSLP by aggregating across sources. The method presented in this section is not specific to piano or IMSLP, and can be extended to other instruments (see Section 3.4), and other music datasets, such as PDMX [7] or audio performances.

2.3.4 Evaluation Subset

To support further evaluation of our music metadata linking strategy, we manually labelled a randomly sampled subset of the graded pieces with their corresponding IMSLP and MB entries. The initial round of annotation found that the graded pieces dataset is dominated by the most popular classical composers (e.g., Bach, Mozart, Bartók, et al.), who are also among the most robustly catalogued. As such, we supplemented the sampling process by first filtering out graded piece entries for which we could find an exact title match or catalogue number match, with the objective of deriving a 'challenging' evaluation set that cannot be trivially solved using regular expressions. After removing some ill-formed entries, this resulted in 450 repertoire list entries.

Each entry was labelled with its 'closest match' in each of MB and IMSLP: the exact match was used if it could be identified, and the next closest match (usually the parent work) if not. If neither of these existed, the 'no match' label was assigned.

Figure 4 shows some of the ambiguities that can occur in the graded pieces. Particularly, this includes cases where the listed piece title has nothing in common with the piece nor movement titles recorded by IMSLP or MB; to resolve such ambiguities during manual labelling, we refer to the listed book. The centrality of the book/edition in the source repertoire listings is a key part of the complication here and might motivate wider ISBN search in future work. Using this process, 447 entries were labelled with the closest MB work and 350 with the IMSLP work; the remaining sampled entries were not labelled as they could not be confidently disambiguated. Table 3 shows the raw label distribution of the evaluation dataset.

By construction, this yields an evaluation dataset that is highly biased towards the 'exact' match label. To better balance the dataset, we constructed final evaluation sets from the manually labelled pairs and inferred relations. For example, if a graded piece entry x is labelled with its exact match y in MB, then we can infer that x is 'part of' $\texttt{parent}(y)$ and 'contains' $\texttt{child}(y)$, using the ground truth parent-child work relations built into MB. Likewise, we can infer that any piece outside this family tree of works is not related to x (and hence 'no match'). As IMSLP lacks parent/child fields, only 'no match' relations are inferred there. The result is a dataset of *(graded piece entry, MB*

	Manually labelled		Evaluation tuples	
Label	**MB**	**IMSLP**	**MB**	**IMSLP**
Exact	394	106	425	99
Part of	21	209	293	190
Contains	6	–	405	–
No match	26	35	477	190
Total pairs	447	350	1600	479

Table 3: Label distribution in the evaluation subset.

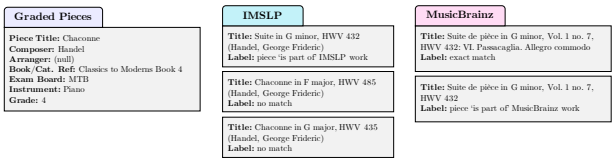


Figure 4: An example of a repertoire list entry from the evaluation set with candidate matches from IMSLP and MB with their ground truth labels. *Chaconne* here refers to movement *VI. Passacaglia of HWV 432* (titled on some recordings as *VI. Passacaile (Chaconne)*); to infer this link, one must generally refer to the contents of the listed book (in this case, *Classics to Moderns Book 4*).

or IMSLP work identifier, ground truth label) tuples; as shown in Table 3, there are **1600** tuples for MB and **479** for IMSLP.

3. RESULTS & DISCUSSION

3.1 Entity Matching for Composers

Though we were able to reconcile most composer values with a WD or MB entity, there were also cases in which these reconciled URIs were not consistent with the existing MB→WD link, as shown in Table 1.

The most frequent cause of failure is when a composer is under-specified (e.g., `Scarlatti` could be Allesandro or Domenico), or when the correct composer is not in the database. In other cases, the composer genuinely cannot be determined without additional information; for example, `Jean Baptiste Loeillet` might refer to `Jean-Baptiste Loeillet of London` (Q1319135) or to `Jean Baptiste Loeillet of Ghent` (Q1347001), both Belgian composers.

OpenRefine's Reconciliation API allows queries to be enriched with additional property values (e.g., date of birth, gender, occupation, ...) to support disambiguation. However, the only additional property provided by the graded piece entries is the free-text piece title, which generally cannot be used for this purpose because the API requires the piece title to first be resolved to a WD entity—which is itself the matching problem we set out to solve.

This is a limitation of our pipeline, which relies on first filtering by composer to make LLM-based piece matching tractable at scale. Future work could explore using LLMs to match composers and pieces jointly, though there are computational drawbacks to this approach.

	MB (n=1600)			**IMSLP** (n=479)		
Model	**Prec.**	**Recall**	**F1**	**Prec.**	**Recall**	**F1**
Qwen2.5-7B	0.88	0.80	0.84	0.95	0.75	0.84
Qwen3.5-4B	0.82	**0.93**	0.87	0.96	0.82	0.88
Qwen3.5-9B	0.90	0.80	0.85	**0.99**	0.68	0.80
Haiku 4.5	0.94	0.87	**0.90**	0.98	**0.87**	**0.92**
Opus 4.8	**0.98**	0.84	**0.90**	0.98	0.84	0.91
Vote-Qwen	0.89	0.82	0.85	**0.99**	0.73	0.84
Vote-All	0.97	0.85	**0.90**	0.98	0.81	0.89

Table 4: Match detection results (positive-class precision/recall/F1) for the evaluation set against MB and IMSLP, across all five base models and both voting ensembles.

	MB (n=1600)		**IMSLP** (n=479)	
Model	**Acc.**	**Macro-F1**	**Acc.**	**Macro-F1**
Qwen2.5-7B	0.46	0.37	0.60	0.40
Qwen3.5-4B	0.55	0.55	0.79	0.60
Qwen3.5-9B	0.57	0.54	0.76	0.56
Haiku 4.5	0.66	0.65	0.85	0.64
Opus 4.8	**0.84**	**0.85**	**0.87**	**0.66**
Vote-Qwen	0.56	0.54	0.78	0.59
Vote-All	0.77	0.76	0.84	0.64

Table 5: Classification (4-way) accuracy and macro-F1 for the evaluation set against MB and IMSLP.

3.2 LLM Entity Matching Evaluation

We evaluate the accuracy of the LLM-based entity matching using the labelled subset of graded pieces introduced in Section 2.3.4. For this, we compare the results of five models chosen to reflect some of the most recent and high-performing options, spanning both open-weight and proprietary models: Qwen2.5-7B [15], Qwen3.5-4B, Qwen3.5-9B [14], Claude Haiku 4.5, and Claude Opus 4.8, along with two majority-vote ensembles: Vote-Qwen (majority vote among the three Qwen models) and Vote-all (majority vote among all five models). All are evaluated using the few-shot configuration described in Section 2.2.

We evaluate two aspects of matching performance. First, *match detection*: grouping the positive labels ('exact match', 'part of', 'contains') into a single category and measuring how often the LLM correctly detects that a match exists at all, regardless of type. Second, *classification accuracy*: the standard four-way accuracy of predicting the correct label ('exact', 'part of', 'contains', and 'no match').

Results are shown in Tables 4–5. The difference between open-weight and proprietary models is less pronounced for match detection than for four-way classification. Among the open-weight models, Qwen3.5-4B achieves the highest detection F1, with scores of 0.87 for MB and 0.88 for IMSLP, whereas Opus 4.8 achieves the highest classification accuracy on both targets (0.84 and 0.87). Neither voting ensemble consistently outperforms its strongest constituent.

3.3 Comparison to Other Piano Difficulty Datasets

Section 2.3 describes how an LLM-based matching pipeline can be used to create a new sheet music difficulty

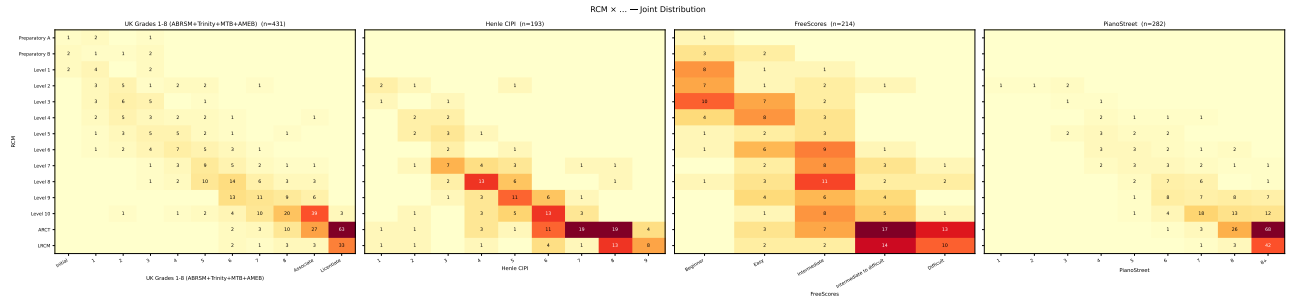


Figure 5: Each cell indicates the number of **MB** works that are assigned the corresponding labels on both axes. A higher RCM grade generally corresponds to a higher UK, CIPI, or PianoStreet grade. 'UK Grades' aggregates entries from ABRSM, Trinity, MTB, and the Australian AMEB (which uses a similar eight-grade system).

	GP	**CIPI** [5]	**PS** [4]	**FS** [4]
# entries	7,737	652	2,816	4,193
# w/ composer	5,057	652	2,699	1,995
IMSLP match	2,389	517	2,100	1,261
MB match	3,304	566	2,000	1,390

Table 6: The number of entries in each query dataset (GP, CIPI, PS, FS) for which the pipeline found a match to an IMSLP or MB work.

dataset, by aggregating exam board-provided difficulty labels with public domain sheet music repositories such as IMSLP. We can hence compare the IMSLP-matched subset of 'graded pieces' (GP) to existing piano difficulty datasets—namely, *Can I Play It?* (CIPI) [5], *PianoStreet-Difficulty* (PS), and *FreeScores-Difficulty* (FS) [4]—by applying the same pipeline to each of these datasets.

As shown in Table 6, the pipeline found IMSLP matches for the majority of CIPI and PS entries, which describe only works in the public domain. Accordingly, fewer IMSLP matches are found for GP, which includes more recent repertoire. The pipeline also failed to match many FS entries to both MB and IMSLP, although we attribute this to poor composer reconciliation, meaning many entries never reached the LLM stage.

Nonetheless, we can use this to compute the 'cross-section' of these datasets—that is, the set of IMSLP or MB work identifiers matched to more than one dataset. Taking the complement of this, we find that our dataset (GP) includes 630 IMSLP work identifiers not shared with matches for any of the other datasets. This suggests complementary matched coverage, although errors in automatic matching prevent us from concluding that these works are genuinely absent from the source datasets.

The cross-section also allows us to compare difficulty labelling scales across datasets and exam boards. Figure 5 shows the overlap between MB works matched to each dataset and those matched to RCM repertoire-list entries, disaggregated by difficulty levels. If the difficulty labels of two datasets or systems were perfectly aligned, we would expect to see the overlap concentrated on the diagonal. This suggests that our derived labels for each MB work are reasonably consistent with publisher-provided labels for CIPI and expert labels assigned by PianoStreet.

3.4 Extension to Other Instruments and the Role of Arrangement

The pipeline described in Section 2.3 matches composers, then compositions (or 'works'); it does not directly attempt to match *arrangements*.

A substantial majority of graded pieces (82.7% overall; 95% for piano and voice; 83% for flute; 82% for violin; 75% for cello; 62% for double bass; and 56% for lever harp) have no arranger credited. This does not guarantee that the IMSLP score represents the same version, so arrangement-level validation remains necessary before extending the pipeline.

4. CONCLUSION & FUTURE WORK

This paper presents a pipeline for reconciling entities across heterogeneous music datasets at scale, combining existing reconciliation services for multiple metadata repositories with LLM-based entity matching. As a concrete application, we use this pipeline to link entries from music exam board repertoire lists to works in IMSLP, supported by MB and WD, to produce a new difficulty-labelled dataset of piano music. We evaluate the models on final sets of 1,600 MB and 479 IMSLP tuples, and find that cross-dataset comparisons suggest broadly consistent difficulty labels and complementary matched coverage.

The arranger-credit analysis identifies voice, flute, violin, and cello as promising candidates for future application of the full pipeline, while highlighting the need for arrangement-level reconciliation.

On the methodological side, LLM-based entity matching is computationally expensive and requires heuristic pre-filtering by composer and embedding-based candidate ranking (top $k = 30$) to make it tractable. Future work might investigate smaller models and report end-to-end retrieval recall.

Richer and more granular metadata in the 'backbone' knowledge graphs (such as WD and MB) would alleviate many of the challenges presented here. That said, there will probably always be unstructured text sources to reconcile to these structured graphs, and therefore a need for flexible matching systems of the kind outlined here.

5. AI USAGE STATEMENT

'AI' usage is a core part of the methodology as described in the main text, and consistent with the ISMIR 2026 AI Usage Policy.

6. REFERENCES

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